Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

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Disclosures and Notes

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AARHUS UNIVERSITY HEALTH

SDCA Steno Diabetes Center Aarhus

Notes on website version

This dissertation is available for viewing online at HeleneThomsen.github.io/dissertation/ or by scanning the QR code in the lower right-hand corner of the title page.

Acknowledgements

Preface

Outline of the dissertation

Papers in the dissertation

Study I

Thomsen, H. B., Lebiecka-Johansen, B., Nørgaard, O., Andersen, T. H., Andersen, S. T., Fagherazzi, G., Hulman, A., & Isaksen, A. A. (2026). Continuous Glucose Monitoring–Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review. JMIR Diabetes, 11(1), e89374. https://doi.org/10.2196/89374

Study II

Thomsen, H. B., Andersen, S. T., Fagherazzi, G., Isaksen, A. A. & Hulman, A. (2026). Foundation model–derived representations of continuous glucose monitoring data: relation to glycemic patterns and cardiovascular risk. (Under review at JMIR AI, (7.30.2026), pending for preprint at JMIR preprints)

Study III

Thomsen, H. B., Li, L. Y., Isaksen, A. A., Lebiecka-Johansen, B., Bour, C., Fagherazzi, G., Doorn, W. P. T. M. van, Varga, T. V., & Hulman, A. (2025). Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes. PLOS Digital Health, 4(6), e0000918. https://doi.org/10.1371/journal.pdig.0000918

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Abbreviations

AGP: ambulatory glucose profile

CGM: continuous glucose monitoring

CV: coefficient of variation

CVD: cardiovascular disease

GMI: glucose management indicator

HbA1c: hemoglobin A1c

MBG: mean glucose

SMBG: self-monitoring blood glucose

T1D: type 1 diabetes

T2D: type 2 diabetes

TAR: time above range

TBR: time below range

TIR: time in range

Introduction 1

1.1 Diabetes

Diabetes is a metabolic disease characterized by high blood glucose levels known as hyperglycemia.[1] The disease is one of the fastest growing global healths challenges.[2] In 2021 it was estimated that 537 million people lived with diabetes, and it is expected to reach 643 million by 2030.[2] The most common forms of chronic diabetes are type 1 diabetes (T1D) and type 2 diabetes (T2D).[1] T1D is an autoimmune disease characterized by the destruction of the pancreatic islet β-cells, leading to an absolute insulin deficiency.[3] Approximately 85%-95% of all people with diabetes in developing countries have T2D.[1] It is the most common form of diabetes and in contrast to T1D it generally begins with insulin resistance in which the body's cells become less responsive to insulin.[4,**Sachs2023?**] Initially, the pancreas compensates by increasing insulin secretion, but over time progressive β-cell dysfunction results in inadequate insulin production to meet the body's demands.[1,4,5] T2D is the most which is is Despite their distinct underlying pathophysiology, both T1D and T2D are characterized by chronic hyperglycemia.

Diabetes can be diagnosed based on a bloodtest to check hemoglobin A1c (HbA1c) levels or through plasma glucose criteria (Table 1.1).[3,**ADA2023_ch6?**]

Although both plasma glucose tests and HbA1c tests are appropriate for diabetes screening, the HbA1c test has several advantages.[3] Some of the benefits are that it requires no fasting and it is less sensitive to stress and changes in nutrition.[3] HbA1c is an indirect measure of blood glucose levels because glucose binds hemoglobin over $\sim$ 120 days lifespan of erythrocytes (red blood cells).[3] However HbA1c testing is not always accessible, affordable or suitable (e.g., for people with anemia or under HIV-treatment).[3] With the guidelines recommendation of using HbA1c, a debate has sparked on the controversy of using a set thresholds as it is well known that racial differences in HbA1c exists.[6] Black individuals have higher HbA1c levels than Hispanic and non-Hispanic White people with diabetes,[3,7] however studies show that the association between HbA1c and risk of complications appear to be similar between ethnic race groups.[3]

1.2 Blood Glucose Monitoring

HbA1c values are also a very important part of diabetes management. The treatment goal is for HbA1c to be below 53 mmol/mol for non-pregnant adults without any hyperglycemia or hypoglycemia (low blood glucose levels). However HbA1c goals should be individualized to fit each individuals specific situation HbA1c can vary from e.g., <48mmol/mol to none.[8,9] There are two tools available for self-assessment of glycemia, the self-monitoring blood glucose (SMBG) device measuring capillary blood with finger-sticks, and the continuous glucose monitoring (CGM) measuring interstitial fluid continuously.[9,10] Both monitoring devices have shown to improve glycemic control in people with diabetes, however the CGM is superior when it comes to immediate information and alerts on asymptomatic hypo- and hyperglycemic events and

Table 1.1: Diagnostic tests for diabetes

Test	Criteria	Description
Fasting plasma glucose	≥7.0 [mmol/L]	Fasting is defined as no calorie intake for at least eight hours.
2-hour plasma glucose	≥11.1 [mmol/L]	Glucose is measured 2 hours after oral ingestion of 75g glucose dissolved in water (OGTT)
Random plasma glucose	≥11.1 [mmol/L]	Individuals with classic symptoms of hyperglycemia (e.g., dehydration and unintentional weight loss) or hyperglycemic crisis (e.g., diabetic ketoacidosis) together with a plasma test taken any time of day.
Glycated hemoglobin (HbA1c)	≥48 [mmol/mol]	

provide a comprehensive glucose profile including both pre- and postprandial blood glucose measurements.[9,11,12]

The CGM devices measures the glucose in the interstitial fluid, which is the fluid around and between cells. It corresponds with plasma glucose although a lag can appear when glucose levels rise or fall rapidly.[**ADA2026_ch7?**] Use of CGM devices has shown to improve glycemic control in both T1D and T2D with improvements in HbA1c levels and more time spent in target range (3.9-10 mmol/L) regardless of age, sex, eduction and income.[**ADA2026_ch6?**,**ADA2026_ch7?**] CGM devices focuses on monitoring and it is possible to get an ambulatory glucose profile (AGP) report from you device (Figure 1.1). The AGP report shows summary statistics over all your data (usually a two weeks period). Some of the most important management measures from the CGM is time in range (TIR), and it is the time spent within normal range in percentage over a given time (often two weeks). Other CGM-derived metrics are time above range (TAR), time below range time below range (TBR), glucose management indicator (GMI), coefficient of variation (CV) and the mean glucose (MBG), number of days CGM was worn and the percentage of time the CGM was active (Table 1.2).[11]

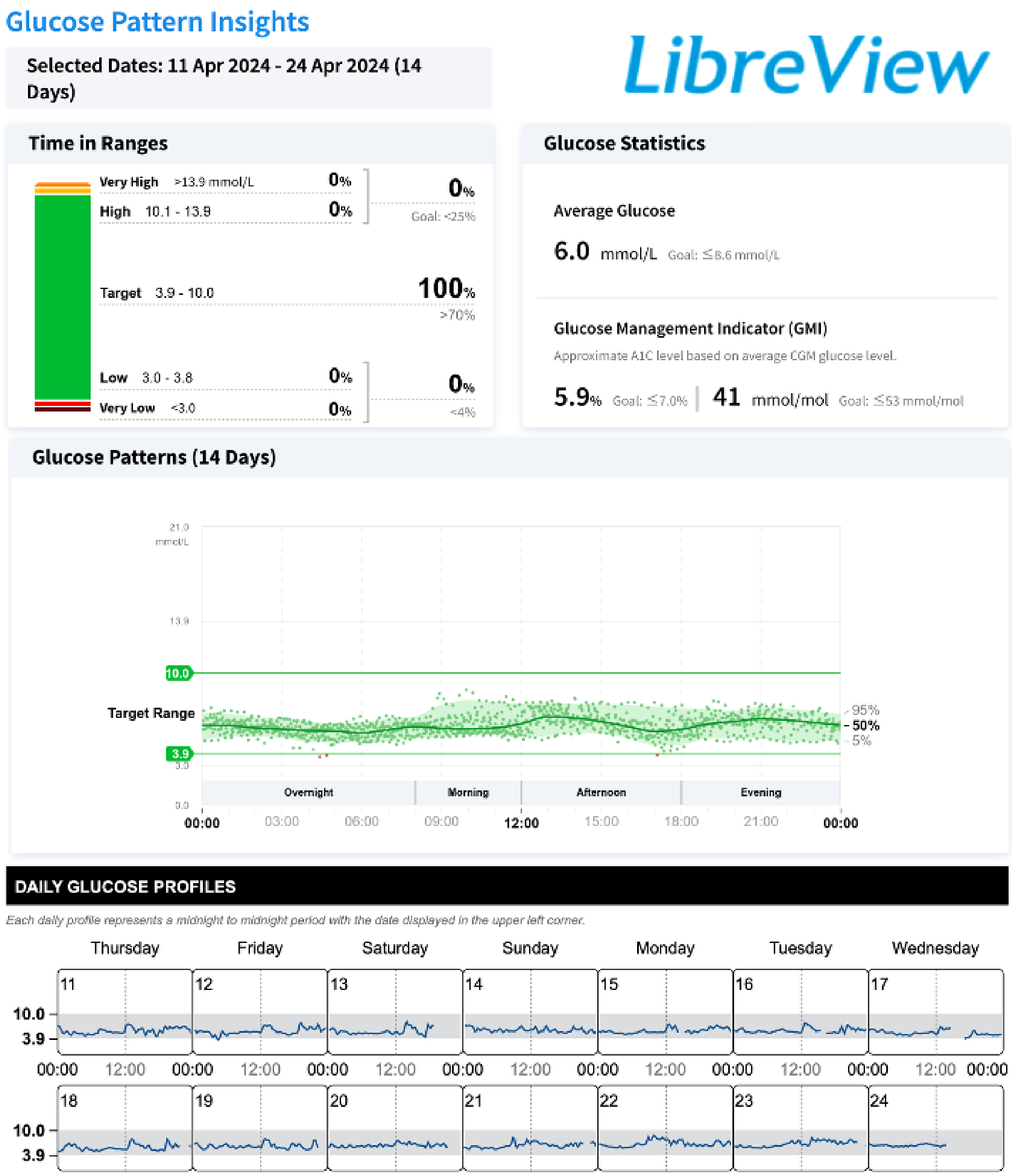

Figure 1.1: Ambulatory glucose profile report from female without diabetes. Data points are from the author herself using an intermittently-scanned (Manuel data upload) FreeStyle LibreLink sensor.

1.3 Management

1.4 Glycemic markers and glucose monitoring

- ☒ HbA1c and its limitations
- ☒ Continuous glucose monitoring (CGM)
- ☐ AIDs use CGMs and algorithms to adjust insulin delivery.[ADA2026_ch7?]

Table 1.2: CGM-derived metrics presented in the ambulatory glucose profile report with target values.

Abbreviation	CGM-derived metric	Range	Recommended targe
TIR	Time in Range	3.9–10.0 mmol/L (70–180 mg/dL)	$> 70\%$
TBR	Time Below Range	< 3.9 mmol/L (< 70 mg/dL)	$< 4\%$
TBR_{54}	Time Below Range	< 3.0 mmol/L (< 54 mg/dL)	$< 1\%$
TAR	Time Above Range	> 10.0 mmol/L (> 180 mg/dL)	$< 25\%$
TAR_{250}	Time Above Range	> 13.9 mmol/L (> 250 mg/dL)	$< 5\%$
GMI	Glucose Management Indicator		No specific target
CV	Coefficient of Variation		$\leq 36\%$

- [ ]Glycemic variability
  - ☐ Metrics of glycemic variability
  - ☐ Clinical challenges of glycemic variability (Think HbA1c and MBG and variability study)

1.5 Cardiovascular diseases and Diabetes

- Cardiovascular complications
- Heart failure in diabetes

Williams2020 expenditures and complications: https://www.sciencedirect.com/science/article/pii/S0168822720301388 ## Cardiovascular risk prediction in diabetes * Traditional risk models * SCORE2-Diabetes risk score * What is SCORE2-Diabetes 10 year CVD risk score? * Limitations of conventional predictors

1.6 Machine learning

Relevant? Artificial Intelligence: The Future for Diabetes Care: https://www.clinicalkey.com/#!/content/playContent/1-s2.0-S0002934320303399?returnurl=https:%2F%2Flinkinghub.elsevier.com%

- Data-driven vs hypothesis-drivenly(knowledge-based) paradigms
- Why machine learning in diabetes and cardiovascular risk prediction?
- Why are ML fantastic for CGM data?
- Types of machine learning methods
  - Logistic regression
  - LSTM and CNNs
  - Foundation models and their advantage
  - The CGMformer is bla bla bla[13]
- Challenges and pitfalls of machine learning in healthcare
  - Overfitting and generalizability
  - Interpretability and explainability

1.7 Fairness and bias in machine learning

- Define race and ethnicity
- Algorithmic bias and group fairness
- Fairness metrics and mitigation strategies
- Trends in the quality of care and racial disparities in Medicare managed care: https://pubmed.ncbi.nlm.nih.gov/16107622/

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Study I: Search strings A

Table A.1: Search in Ovid MEDLINE (Search date: 22 January 2024)

<table>
<tr><th>#</th><th>Searches</th><th>Results</th></tr>
<tr><td>1</td><td>Blood Glucose Self-Monitoring/</td><td>10241</td></tr>
<tr><td>2</td><td>Continuous Glucose Monitoring/</td><td>34</td></tr>
<tr><td>3</td><td>(((blood-glucose or bloodglucose or glucose) adj2 (monitor$ or selfmonitor))orcgmorbgmoriscgmorrtcgmor(timeadj3range)).ti, ab, kf, ku or bloodsugar$ or glucose$ or glyc?emic$ or sugar).ti, ab, kf, kw.)adj5(flashorfluctuat or intermittent$ or scan$ or variabilit).ti, ab, kf, kw.|8669||5|or/1−4|38366||6|expCardiovascularDiseases/|2760764||7|(cardiovasc or cvd or cardiometabol$ or cardio-metabol$ or heart$ or macrovasc$ or vasc).ti, ab, kw, kf.|2182473||8|or/6−7|3971342||9|expDiabetesMellitus/|518864||10|(diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d).ti,ab,kw,kf.</td><td>815092</td></tr>
<tr><td>11</td><td>or/9-10</td><td>875527</td></tr>
<tr><td>12</td><td>and/5,8,11</td><td>2416</td></tr>
<tr><td>13</td><td>exp Animals/ not Humans/</td><td>5189493</td></tr>
<tr><td>14</td><td>12 not 13</td><td>2300</td></tr>
<tr><td></td><td>Updated search: 11 March 2025</td><td></td></tr>
<tr><td>15</td><td>limit 14 to (da="20231201-20261231" or dt="20231201-20261231" or ez="20231201-20261231" or ed="20231201-20261231")</td><td>310</td></tr>
<tr><td>16</td><td>or/14-15</td><td>2610</td></tr>
</table>

Table A.2: Search in Embase (Ovid) (Search date: 22 January 2024)

<table>
<tr><th>#</th><th>Searches</th><th>Results</th></tr>
<tr><td>1</td><td>blood glucose monitoring/</td><td>37574</td></tr>
<tr><td>2</td><td>exp continuous glucose monitoring system/</td><td>5648</td></tr>
<tr><td>3</td><td>(((blood-glucose or bloodglucose or glucose) adj2 (monitor$ or selfmonitor))orcgmorbgmoriscgmorrtcgmor(timeadj3range)).ti, ab, kf, kw.|4850 or bloodsugar$ or glucose$ or glyc?emic$ or sugar).ti, ab, kf, kw.)adj5(flashorfluctuat or intermittent$ or scan$ or variabilit).ti, ab, kf, kw.|16430||5|or/1−4|81510||6|expcardiovasculardisease/|5122163||7|(cardiovasc or cvd or cardiometabol$ or cardio-metabol$ or heart$ or macrovasc$ or vasc).ti, ab, kw, kf.|3044925||8|or/6−7|6303633||9|expdiabetesmellitus/|1239184||10|(diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d).ti,ab,kw,kf.</td><td>1237019</td></tr>
<tr><td>11</td><td>or/9-10</td><td>1501812</td></tr>
<tr><td>12</td><td>and/5,8,11</td><td>11217</td></tr>
<tr><td>13</td><td>(rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset$1).ti. and animal experiment/</td><td>1237966</td></tr>
<tr><td>14</td><td>animal experiment/ not (human experiment/ or human/)</td><td>2601109</td></tr>
<tr><td>15</td><td>or/13-14</td><td>2672240</td></tr>
<tr><td>16</td><td>12 not 15</td><td>10861</td></tr>
<tr><td>17</td><td>limit 16 to exclude medline journals
Updated search: 11 March 2025</td><td>2176</td></tr>
<tr><td>18</td><td>limit 17 to (dc="20231201-20261231" or dd="20231201-20261231" or rd="20231201-20261231")</td><td>467</td></tr>
<tr><td>19</td><td>or/17-18</td><td>2643</td></tr>
</table>

Study I: Included and excluded reports B

C Study I: Geographical location

Asia (n=27) • China (n=19) • Japan (n=8) Europe (n=22) • Italy (n=7) • 3 Belgium (n=3) • 2 Ukraine (n=2) • 2 Finland (n=2) • 2 Spain (n=2) • 1 Greece (n=1) • 1 Malta (n=1) • 1 Sweden (n=1) • 1 The Netherlands (n=1) • 1 Portugal (n=1) • 1 Russia (n=1) North America (n=2) • 2 USA (n=2) Australia / Oceania (n=1) • 1 Australia (n=1)

Borg 2011(ADAG study) had study centers in: • Europe: Netherlands, Denmark, Italy • Africa: 2011 • North America: USA

Borg R, Kuenen J C, Carstensen B, Zheng H, Nathan D M, Heine R J, et al. HbA1(c) and mean blood glucose show stronger associations with cardiovascular disease risk factors than do postprandial glycaemia or glucose variability in persons with diabetes: the A1C-Derived Average Glucose (ADAG) study. 2011; ADAG Study Group, editor. Diabetologia. 54: 69–72. doi:10.1007/s00125-010-1918-2

Study I: Subclinical results D

- ☐ Add table with subclinical study design overviews
- ☐ Add table overview of subclinical results

E Study II: Prediction stability

Prediction stability plots (**?@fig-pred_stability**) revealed more stability the fewer predictors.

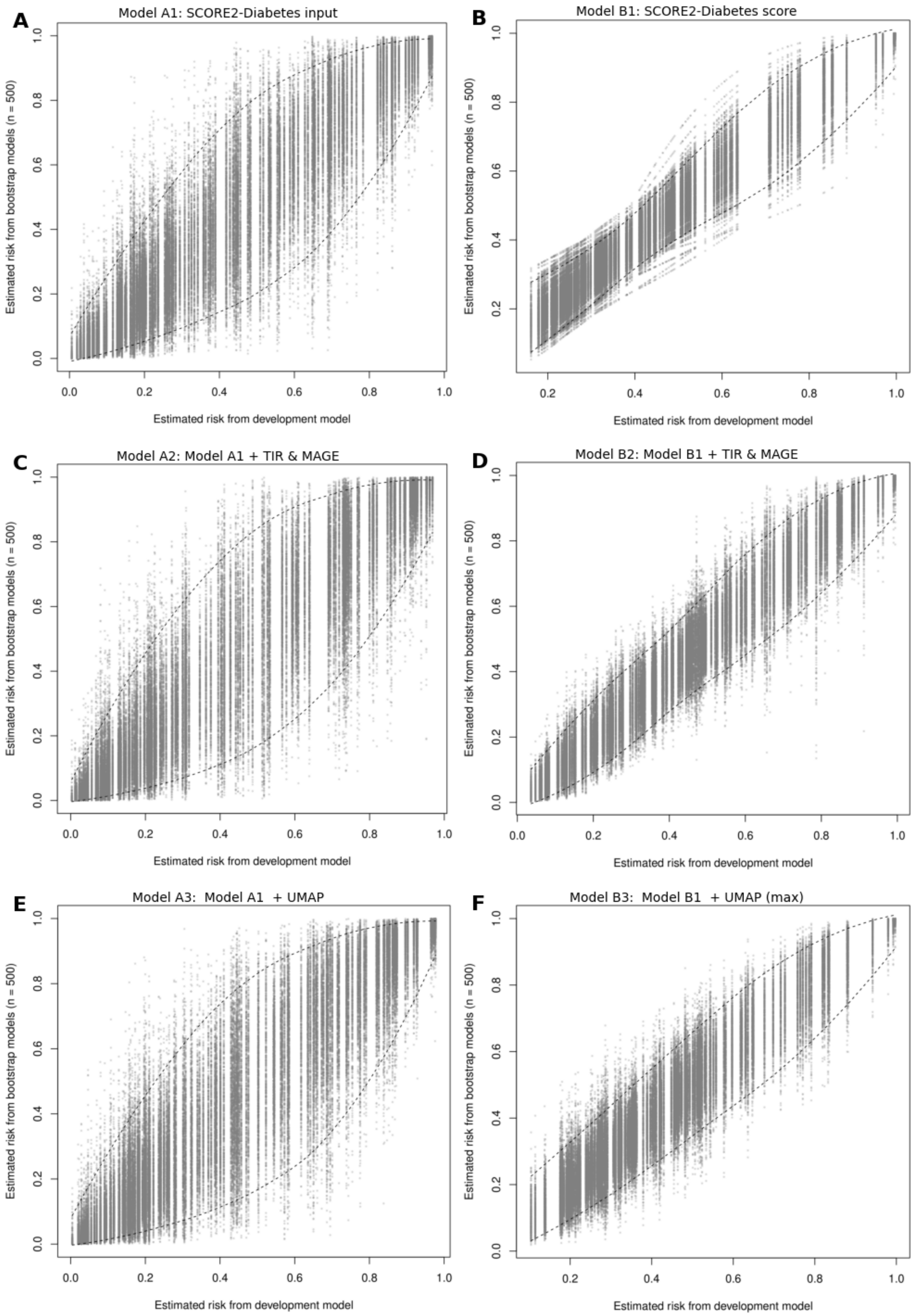

Figure E.1: Prediction stability plots show the predictions (estimated risk) for the full dataset from the development model (equivalent to the apparent model) on the x-axis, and the predictions from all bootstrap models on the y-axis. Tight alignment along the diagonal indicates higher prediction stability.

F Study III: Surveillance Error Grid Results

Table F.1: Results from the Surveillance Error Grid for White participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
White	LOCF	None	69%	69%	69%	69%	69%	69%
		Slight	26%	26%	26%	26%	26%	26%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	28%	28%	28%	28%	28%	28%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Generalized	None	72%	73%	73%	73%	73%	73%
		Slight	25%	25%	25%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Transfer Learned	None	72%	72%	73%	73%	73%	73%
		Slight	25%	24%	24%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

☐ maybe add this one to appendix, since I think they show the same things

Table F.2: Results from the Surveillance Error Grid for Black participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
Black	LOCF	None	71%	71%	71%	71%	71%	71%
		Slight	24%	24%	24%	24%	24%	24%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	27%	27%	27%	27%	27%	27%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%

Group	Model	Zone	0	20	40	60	80	100
		Extreme	0%	0%	0%	0%	0%	0%
	Base -Generalized	None	75%	74%	75%	75%	74%	75%
		Slight	23%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Transfer Learned	None	75%	74%	75%	75%	74%	75%
		Slight	22%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

G Full papers

G.1 Paper I

G.2 Paper II

G.3 Paper III

Addenda and errata H

This chapter contains an overview of:

- Addenda: contents added since the submitted version of the dissertation.
- Errata: corrections to errors discovered since submission of the dissertation.

H.1 Addenda

H.2 Errata